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#Hamming distance := number of positions where two strings of equal length differ
#Here string length is guaranteed 2

.globl main
.text

main:

#get string 1
li $v0, 4
la $a0, s1
syscall
li $v0, 8
la $a0, str1
li $a1, 10
syscall

#get string 2
li $v0, 4
la $a0, s2
syscall
li $v0, 8
la $a0, str2
li $a1, 10
syscall


#comparing
lb $t1, str1
lb $t2, str2
sne $t3, $t1, $t2

lb $t1, str1+1
lb $t2, str2+1
sne $t4, $t1, $t2

add $t0, $t3, $t4
move $a0, $t0
li $v0, 1
syscall

# exit program
li  $v0, 10         # system call code for exit = 10
syscall             # call operating system

.data
s1: .asciiz "String 1:"
s2: .asciiz "String 2:"
str1: .space 10
str2: .space 10
```